

Information Management in the Emerging Edge

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Virtual Event
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NIST

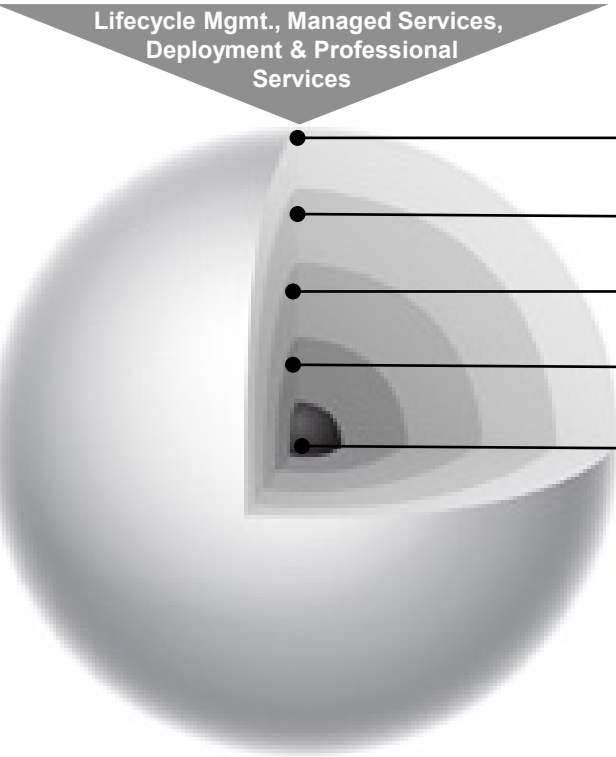
**NAMED DATA
NETWORKING
COMMUNITY
MEETING**

October 28
October 29
2021

DELLTechnologies

Edge Definition and Stratification

Lifecycle Mgmt., Managed Services,
Deployment & Professional
Services

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- Strat 4 – Devices, Sensors, Vehicles, Physical (enclosure, power, env, security)
 - Strat 3 – Infrastructure Compute, Network, Storage, Security
 - Strat 2 – M&O, Virtualization, Cloud Native Control Plane
 - Strat 1 – Application IDE/Mgmt., Services, API, Data, Security Management Stacks
 - Strat 0 - Applications and Data

Edge is a distributed system consisting of scalable infrastructure with management and orchestration software that enables applications and services to:

- Execute control and data insight with low latency.
- Ingest and process data near its source.
- Maintain governance and compliance with data locality and privacy

The Edge system will operate in a constrained environment and provide cost-effective, secure, high value capabilities and performance across multiple verticals.

Edge is not a single device, place, thing, vertical specific or a public cloud in a different form factor.

Site Acquisition,
Construction, Site Ops

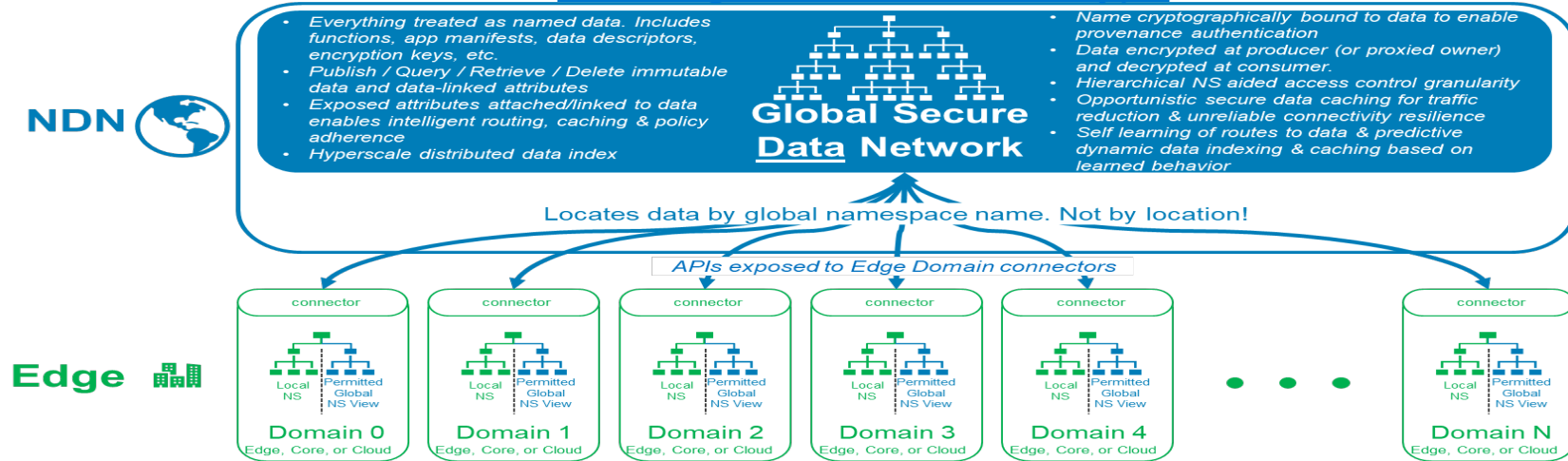
The Key Challenges of Edge Information Management

1. Data will be transitory to edge (Ingest, processing, inference and analytics).
2. Autonomous systems will rely on edge-based computation and support.
3. Assumption of data locality is not guaranteed for Edge.
4. Assumption of data reachability is not guaranteed for Edge.
5. Data visibility is problematic for many Edge workloads.
6. Data provenance and security is needed.
7. IP multicast/unicast needs significant overlay capabilities to operate effectively.
8. The edge will include mobile workloads.
9. IT based catalogs data pipelines poorly suited to the polydirectional high scale Edge.
10. Cloud solutions focus on storageaaS, Analytics and AI enablement.
11. Most Orchestration frameworks do not consider data in many decisions.

NDN Advantages in Dell Research

- Data naming can be automated and provide semantic information can be abstracted and managed without application knowledge.
- Inherent ability to perform distributed dataset synchronization (SVS).
- Can inform a movement policy that can be control by application “shim”

Dell Edge Research Prototype



NDN Ideas for Future Evolution

- Interest Flooding – Ability to send “ranged” interest request with single interest packet.
- High performance NFD.
- Continued development of State Vector Sync.
- Kubernetes with a Container Storage Interface for NDN.
- Improved and modernized set of development tooling/environment:
 - Improved network deployment tools (setting up complex network topologies is cumbersome/error prone).
 - Telemetry/Metrics tool integration (e.g. Prometheus)
 - NFD formal plug-in architecture (many add-on diagnostic tools are applications that run together with an NFD (e.g. ndntrace), these have to be managed/monitored/maintained. Having a formal plug-in architecture would make management much simpler (though at the potential risk of making the NFD less stable)

The logo for Dell Technologies, featuring the word "DELL" in a stylized font where the "E" is composed of three slanted parallel lines, followed by the word "Technologies" in a clean, sans-serif typeface. The entire logo is white and centered against a solid blue background.